

# CSC 4510 / 6510 – Automata Syllabus

Course Prefix, Number, and Name: CSC 4510 / 6510 – Automata

Semester and Academic Year: Fall 2026

## 1 Time

2:45pm–4:30pm on Tuesdays and Thursdays

## 2 Room

Classroom South, Room 205

## 3 Instructor

- **Name:** Murray Patterson
- **Office:** 1 Park Place, Room 948E
- **Email:** mpatterson30@gsu.edu
- **Zoom:** <https://gsu-edu.zoom.us/my/mpatterson30>

## 4 Office Hours

- Wednesdays from 2:30pm–4:30pm in my office or via Zoom (I will be online during this time)
- After class: I will remain in the classroom
- By appointment: please discuss with me or send me an email to arrange a time

## 5 Teaching Assistants (TAs)

- **Name:** Avais Jan
- **Email:** ajan3@student.gsu.edu

## 6 Prerequisites

CSC 2720 with a grade of C or higher. Students must also meet the Computer Science Major Eligibility Requirement in order to enroll in this course.

## 7 Reference Materials

Notes and other online resources will be provided as the course proceeds.

## 8 Course Content

Georgia State University iCollege — <https://icollege.gsu.edu>

## 9 Course Overview

Welcome to CSC 4510 / 6510 at Georgia State University! This course concerns the theory of computing devices and the languages they recognize. It will cover deterministic and nondeterministic finite automata, context-free grammars, pushdown automata, Turing machines and undecidability.

## 10 Course Structure

- **Lecture** — Classes will be conducted in a traditional lecture format
- **Homework** — Assignments will build on the lecture content
- **Discussions** — Opportunities to share questions about the course material, homework, etc.

## 11 Exams

There will be a midterm and a final exam in this course. The exams will involve questions and problems which cover lectures, homework assignments, and readings.

## 12 Grade Scale

| Grade | Point Equivalent |
|-------|------------------|
| A+    | $\geq 97$        |
| A     | $\geq 90$        |
| B+    | $\geq 87$        |
| B     | $\geq 80$        |
| C+    | $\geq 77$        |
| C     | $\geq 70$        |
| D     | $\geq 60$        |
| F     | $< 60$           |

## 13 Grading (*subject to change*)

- Homework Assignments (20%)
- Midterm Exam (40%)
- Final Exam (40%)

## 14 Course Schedule (*subject to change*)

1. Regular Languages, Finite Automata
2. Context-Free Languages, Pushdown Automata
3. Recursive and Recursively Enumerable Languages, Turing Machines
4. (Time permitting) Other Models of Computing: Lambda Calculus, Recursive Function Theory, Random-Access Machines

## 15 Make-up Policy

### Exams

There are no make-up exams unless the student missed the exam due to a pre-arranged excused absence *e.g.*, participation in a GSU sports event, observance of a religious holiday (see <https://belonging.gsu.edu/religious-observances/>), or an emergency, etc. In all cases, documentation needs to be provided before or after, *e.g.*, a note from the coach, a mention of the religious holiday, or a slip from the doctor, etc. Only official excuses will be accepted. **Any uncoordinated, unexcused missed exam will result in a score of zero for that exam.**

### Homework

Each homework assignment is due at the beginning of class on the due date.

## 16 Academic Integrity Statement

Academic dishonesty is a serious violation of the trust upon which the success of our University depends. Cheating and plagiarism can not only result in a poor grade and penalties from the University, but it can cause your mentors and peers to mistrust you and could keep you from developing the habits to make you a successful student and a successful worker in your future career. The University's policy on academic honesty is published in the Student Handbook, <https://deanofstudents.gsu.edu/student-conduct/#academic-honesty>, and includes dishonest actions such as cheating, plagiarism, and facilitating academic dishonesty. Please be aware that violations of this policy will result in a grade of "F" for the assignment or the course. If you have any questions about the policy or are unsure if something you are about to do counts as academic dishonesty, please come to my office hours and we can discuss it.

## 17 Course Evaluations

Your constructive assessment of this course plays an indispensable role in shaping education at Georgia State. Upon completing the course, please take the time to fill out the online course evaluation.

## 18 Extended Absences

For students, the Dean of Students' Office will continue to provide faculty with notifications when students file **Professor Notification of Absences (PNAs)**. This notification indicates that the Dean of Students Office has reviewed the documentation related to a student's medical circumstances. For more information about this, and how to submit such a notification, see <https://deanofstudents.gsu.edu/student-assistance/#professor>.

## 19 ACCESS AND ACCOMMODATIONS POLICY

Students who wish to request accommodation for a disability may do so by registering with the Access and Accommodations Center. Students may only be accommodated upon issuance by the Access and Accommodations Center of a signed Accommodation Plan and are responsible for providing a copy of that plan to instructors of all classes in which accommodations are sought. The Access and Accommodations Center may be contacted at 404-413-1560 or by email at [access@gsu.edu](mailto:access@gsu.edu). Any student who may require special accommodations is requested to make an appointment with the instructor at the beginning of the semester. Students must self-identify so that arrangements can be made according to university policy.

## 20 Basic Needs Statement

Any student who faces challenges securing their food or housing and believes this may affect their performance in the course is urged to contact the Dean of Students for support. Furthermore, please notify the professor if you are comfortable in doing so. This will enable us to provide resources that we may possess. The Embark program at GSU provides resources for students facing homelessness and Panther's Pantry provides resources for students facing food insecurity.

## 21 Disclaimer

The course syllabus provides a general plan for the course — deviations may be necessary.